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H8SCOOP

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by Henry Fale

BITS & PIECES

This issue of H8SCOOP should look different to you. I have prepared this issue using SPELLBINDER word processor under CP/M, with my H89, and printing on the Diablo 630 printer in the Proportional Space mode. I have installed the Magnolia Microsystems Double Density disk controller board and am using the Tandon drives which give me about 750K storage per drive. I can do a lot of storing and editing on a single drive system!

The implemented improvements such as Proportional Space and printing 7 lines per inch instead of the normal 6, means in this issue you are getting between 30 and 40% more information within these pages than last month. In other words, this 8 page issue this month is about equal to 12 pages using last month's format, all without additional cost to you, my valued subscribers. I am just learning this word processor, and I figure it will take a few months to iron out all the bugs, so I ask you to be patient with me while I change my whole system and thinking over to a more efficient operation. Please let me know your comments on this format, or forever hold your peace!

I am thoroughly impressed with this word processor, and in my opinion, it BEATS anything else out there, while being simple to use, a very rare combination indeed. In fact, I like this so much (it supports special features of the H19/Z89 terminals) I may start to distribute and support it. More on that latter.

400K DRIVES STILL GOING STRONG

I have sold over 150 of the Tandon drives to date, and because of my volume, I was able to get a slightly reduced price. This means I will not be increasing my price as was reported last month in H8SCOOP. I will hold this price for awhile anyway. The cost of the drive with the software will remain at \$495, with additional drives going for \$465, or single drives purchased as the first drive without the software will be \$470. Cabinets are still available. Don't forget, that price includes TOTAL SUPPORT, not just the drive and "see you around". This is important to note, especially in a situation such as this with so much totally new involved. Any problems will be supported by myself, and the owners of the 150 drives will testify to that. I have already had reports that my documentation is more complete than that supplied by others, and that is understandable, as I have added my own experience to the documentation, and continue to update it on feedback from other users.

If you want to get up and running fast with 400 or 800K, stick with QUIKDATA, the parent corporation of H8SCOOP, and I won't let you down.

One last point about support. Don't buy the drives from someone else, especially the MPI, because you got a "good deal", and then buy the software from someone else (or better yet, copy it from a friend), and then come to me for support--NO DICE! If you want to support others, fine, do so at your own risk. I have a reputation at stake, and have put it on the line from the start in this endeavor. I offer and will continue to offer TOTAL SUPPORT for standard systems. As far as the DG systems, or anyone running anything else at 4MHz, I strongly suggest you return to 2Mhz for the time being. The driver was written for 2 not 4 Mhz, and there have been problems. The new release will hopefully incorporate the 2 and 4 Mhz options, but till then, you're asking for troubles.

I now have a DUMP program that works with this drive, under HDOS, and there is reportedly a DUP coming. DG Electronics, POB 1124, Denison, TX 75020 has the DUMP available for \$49.95 called PRO-DUMP that will support this drive, and they should shortly have a DUP. This PRO-DUMP,

by the way, is a part of a package called "DISK MANAGEMENT UTILITY PACKAGE", which also includes a Block Scan Utility (BAD), and more.

I have also had reports that individuals have installed this drive IN the H89 with no problems. It seems that double track may work OK, but double density is likely to give more problems. If you have wanted to mount this drive in your H89 but were afraid to act, try it, it looks like it may work.

I have had reports that using the new SY.DVD driver some of the DUP type programs will not run even using the standard discs. Someone at HUG found the answer to that. SET SY: SELECT 0 and you should have no problems--it worked for me.

Here's something from the HUG.BB from Bill Moss. "Those using single-sided, single density, 40-track disks with the Tandon 400K drives should be cautious as I have disks which will pass the limited media check of the Ultimeth SY: driver but will turn up with hard read errors. I suspect this is due to not checking each bit, or from poor magnetic retention quality on the flip side of the disk. All the programs I have had are with Verbatim disks that are about two years old (but used very little). No problems with the newer "Datalife" Verbatim product or with any Memorex disk.

"A further word of caution is that as more high density 5" drives are sold, it seems likely that the ability to use the SD-SS-40T disks in the 400K drives will probably diminish, since those that fail the production testing for high density capacity will be labeled for low density use. As it is now, most disks are probably only tested for SDSS capability and probably are good at higher density also."

Another note from Bill from the HUG.BB that may be of interest "One problem with the Tandon drives and all of the present and future high capacity drives is the sectors lost via large number of sector/cluster. With the Tandons at 400K a file takes up disk space in blocks of 8. Thus, a one sector file is wasting 87.5% of the allocated space (this will not be the general case, however. If you have purchased this drive and plan to fill it up with 1 sector files, you missed the boat, but Bill's point is valid and well taken--ed). I have written a program to pack an entire disk into a single file and restore it with flags set as on the original disk. For example, if one copies the HDOS distribution "SOFTWARE TOOLS" disk to a 400K drive, 634 sectors are consumed; when packed however, only 340 sectors are used. I am currently arranging marketing, but it should be available with source code included within the month for under \$20 (this was dated 8/8/81--ed). If interested in this, contact Bill Moss/1507 Riverview Ln/Bradenton, FL 33529 or I'll keep you posted through H8SCOOP or the 400K user newsletter--once I get it going.

Refer to REQUEST section for some info on 8" H8 controller for CP/M.

For any interested, I just got hot off the press a limited number of service manuals for the TANDON TM-100-4 drives and will sell them for \$25 each plus \$3 shipping and handling. I have 3 now and will be getting more.

ONCE MORE ON FLIPPING DISCS!

I thought I said the last, but I want to quote a paragraph from Microcomputing, August 1981 page 44 from an article on 800K floppys. "Diskette manufacturers warn against the practice (flipping the disc and using the other side--ed), however. As a diskette is used, the fiber layer inside collects dirt and dust and holds onto it. If the diskette is turned around, it now turns in the opposite direction, and hence much of this dirt and dust is released. They warn that this could

result in increased disk and head wear as well as data errors."

Let me add this. Don't anyone who purchased the Tandon drives from me use any of these "flipped" discs in the drive, as the heads are more delicate. I will not support drives where head damage shows this may have been the case, and I've already had two!

SOURCES FOR STATIC RAM's

Peter Ruber writes in to inform us that the 16K RAM's used in Heath's 16K memory board are available from QUEST ELECTRONICS/POB 4430/Santa Clara, CA/95054. Their 2114 Static Rams sell for \$3.75 to \$4.25 depending on the speed rating, which is about half of the \$7.68 Heath charge. He says QUEST has a top-notch 82 page catalog of computer related parts, boards, hardware, etc. Their service and warranty policy is good. Quest, along with Netronics Corp, are the two principal suppliers of RCA 1802 system computers. Pete also writes in that another good source of parts is DIGI-KEY corp/Highway 32 South/POB 677/Thief River Falls, MN 56701. They issue catalogs every two months containing mostly computer-related parts. Thanks for the feedback Pete!

Milt Krauthoff found a source of the 4K X 1 static RAM equiv. #4044 for \$4.50 each. These and other IC's are available from Hanley Engineering/13400 Northup Way Bellevue, WA 98005.

ON MEMORY AND H9 MODS

FRANK PFEIFER writes in that info from TRIONYX on the Dynamic RAM chips for Z80 leads him to go to 4116 chips with a speed rating of 150 ns for running Z80 at 4Mhz, as the 200 ns may be marginal at 4 Mhz. Trionyx also informed Frank (and I heard this myself) to stay away from American made chips and go for the Japanese RAM's such as MITSUBISHI. These are available from JDR Microdevices, Inc, 1101 S. Winchester Blvd, San Jose, CA 95128. The RAM is 150 ns, made by Toshiba, plastic case, and cost \$21.95 for 8 as of this printing.

Frank also has some good comments about NORTHWEST COMPUTER DEVICES, the outfit that has all the neat H9 mods. He says "I purchased a PC board from them for the 24 line conversion and an update of the INTERFACE AGE article. Careful planning of where to cut on the boards and wiring layout produced a 24 line conversion without any problems. I tried to use some of those extra existing holes in the PC traces as much as possible. Otherwise I took a number 57 drill and drilled through the PC board just bumping the trace I want to connect to. I also did not cut traces. I prefer reaming out a point on the trace with a small drill.

"The FLICKER FREE mod allows nice screen activity at 4800 baud. For Cursor Control, I was given information on how to delete the cursor altogether by using one of the control lines. The original ERASE PAGE mod from HUG info which used CTRL-E. I simply switched the lead from ERASE PAGE to the ERASE EOL switch. Now a CTRL-E will allow erasing from a character position to the end of a line.

"I was further provided with a mod to blink the cursor. The blinking eventually became bothersome and is not like the blinking cursor on my APPLE II. On the APPLE, the cursor is internally programmed to appear only on inputs; on the H9 it is 'always present'. However, with the ability to delete the cursor altogether under software control (after making the mod), you can call up the cursor when you want to. This also helps the GRAFIX on the screen too, as you don't have a cursor ever-presently floating around.

"The GRAFIX has two minor flaws. On my system, whatever is printed in the first screen column has extra video data enabled and extraneous pulsating dots in the last column. (If you don't print the first column, everything is fine). The other is that GRAFIX does not enable the same blanked dot columns on the three used for inter-character spacing. So, if you're a perfectionist, a GRAFIX character and a regular character would be off from each other by 3 dots."

"The key top set and four arrow key replacement probably is the more "obvious" worth while change. No "stick-on" labels (except for some of the special function H9 keys). In fact a friend of mine thought I bought a new terminal." For more info on this, you can contact Frank at 4136 Park Av, Brookfield, IL 60513.

MORE ON H9 MODS

Tor Tverre, Monradsgate 3 A, Oslo 5, NORWAY has installed numerous mods to his H9 including a new keyboard. He welcomes anyone interested in what and how he has done it to write to him for information. This, by the way, was sent in May 1980 and is first getting printed now, giving you an idea of the information I have to sort through to come up with a newsletter!

Some of the modifications Tor has installed in his H9 include a new keyboard with numeric pad, Upper and lower case, 24 line display, graphics with a resolution of 160 X 72, and Addressable cursor with both Home and Erase Page software control.

For the Keyboard, Tor used the George Risk KB 756 and the 710 numeric pad. He says the quality of the keyboard is only fair, but far superior than the original one. The Upper and Lower case mod is similar to that in REMark #3, but with different logic. The 24 line mod is not difficult, although time consuming and must be done carefully. It appears NORTHWEST now has a kit for that which simplifies things.

I print this info, although the H9 is not too much around anymore, because I have been getting lot of request on what to do with the H9's sitting in the closet! It seems NORTHWEST COMPUTER SERVICES, 8503 N. E. 30th Ave, Vancouver, WA 98665 has about the most to offer for the H9, with lot of experience, since they have been supporting the H9 for many years now. Just an idea of prices are as follows: GRAFIX display for the H9 \$59.95 kit, Eight function cursor control mod--\$29.95 kit, Flicker free 4800 baud operation mod--\$69.95 kit, Lower case entry to lower case characters from keyboard mod--\$28.88 assembled, Lower case display to improve readability--\$34.77 assembled, and set of contour key caps with molded in alphanumeric is \$12.50 for the set.

COMMENTS ON PASCAL LANGUAGE

Fred Pospeschil compiled some data on various PASCAL's verses BH BASIC. He did some simple benchmark tests on the H8 with front panel running (takes about another 10% processor speed). Here are the results in table form.

LANGUAGE	TIME(seconds)
Pascal MT+	6 *
C80	8
Tiny Pascal	10
UCSD Pascal	31
Lucidata Pascal	45
B.H. BASIC	180

You can see from the table that ALL are considerably faster than BH BASIC. Fred adds "I have completed two graphics programs using Tiny Pascal and find it highly satisfactory for applications which do not require floating point, trig, and complex data structures. It's small, fast and the PORT command is handy when working with the color graphics board. (Fred and others have some nice graphics stuff available for the color graphics board under Tiny Pascal, which HUG now distributes).

*The pascal MT+ compiler is progressing well and is about 90% converted to HDOS. I suspect that its speed should be in the ballpark with C80 and Tiny Pascal. An East Coast user of MT+ under CPM reports that it runs about five times faster than UCSD which would make the test program run in about 6 seconds. For what it is worth, I have had several reports that the MT+ Pascal is "the best software development tool available" and that it performs tasks at about half the speed of a properly structured assembly language program.

"I now have a copy of Lucidata Pascal and find it a highly cost effective package. It can be run on a single drive system however using STAND-ALONE allows for creating bigger programs (so does the 400K drives which it works with--ed). Its random file access is the best I have seen and manipulating the CRT and keyboard is quite simple. The current release lacks built-in trig functions but this does not hinder our development of a commercial order processing support system. At present, the trig functions are available as procedures but may be included in the compiler under version 3.0. The support from POLYBYTES (325 19th St. S. E., Cedar Rapids, IA 52403) has been prompt and outstanding." You can purchase this from Polybytes for \$95, or \$65 with group orders of 5 or more, an excellent deal (ed)!

Fred tells me "PASCAL PROGRAMMING STRUCTURES" by George Cherry (Reston Publishing Co., 11480 Sunset Hills Rd, Reston, VA (703) 437-8900) is an excellent introductory text. I have also found PASCAL by Paul M. Chirlian, from Matrix Publishers, Inc, 30 NW 23rd Pl, Portland, OR, to be a good book, and am going through the Heath PASCAL course. The sample programs and materials provided by POLYBYTES is also a great help--ed.

One great point of the Lucidata Pascal distributed by Polybytes is it runs under HDOS, which means it will work with the 400K Tandon drives. I have been using mine with this Pascal for some time now. Most Pascal's do not run under HDOS, so 400K is nix! Another excellent feature is the random file operations which most Pascal's do not support. I have been meaning for some time now to start a Pascal corner in this newsletter using the Lucidata Pascal, but haven't got to it yet. I talked to Bob Ellerton of HUG (really nice guy) and we agreed that I would do the column for REMark magazine, so be looking for it in a few months.

CASSETTE SOFTWARE CONTEST

In an effort to help attract the cassette system owners in the Baltimore area, the local User's Group is sponsoring a software contest. Software will be accepted at the Townson Heath Store with a \$5 entrance fee. All entries must be received by October 1. At the end of the contest period the entries will be judged by an impartial group at the store. The winner will receive all the entrance money (No money will be retained). All tapes will be placed in a common software library at the store, and will be available to everyone who wants it free of charge. They will have to make their own copies. This contest is for CASSETTE programs only. Anyone interested in this contest or in forming a separate cassette group may call William Frey, or the store manager. Anyone interested in joining our disk group may call also. As of this point in time there are no membership fees, so you have nothing to loose and everything to gain by joining. The address: , Baltimore Heath User's Group, 6106 Marlora Rd, Baltimore, MD 21239 (301) 323-6093 after 6PM.

NEW STUFF

ON PC-12 FROM ENVIORNMENTAL CONTROL SYSTEMS

A new and improved Ni-cad battery for the PC-12 clock/12 parallel port board is available for owners of the PC-12 for \$10 postpaid. The average life for the Nicad installed on most boards seems to average about a year. The life of the new Ni-cad should be at least 5 years and is highly recommended. Release 2.0 of the support disc has been delayed and will be shipped shortly. This note from Donald Long of Enviornmental Control Systems, 9319 Willowview Ln, Houston, TX 77080.

ON H14 RIBBONS FROM PC & J GRAPHICS

I've been asked to add a note to the info in last month's H8SCOOP on the better H14 ribbons from PC & J Graphics Co, POB 108, Deer Park, NY 11729 (516) 667-8076. Please make note that if your are from New York, you must add the appropriate sales tax when placing the order. I usually don't mention that since I feel by now everyone should know there is

tax involved when buying anything state locally.

CP/M VERSION OF TOUCH TYPIST

Ron Rocheleau of Newline Software, POB 402, Littleton, MA 01460 tells me the CP/M version of TOUCH TYPIST is now available. It is identical to the HDOS version (see page 3 JUNE H8SCOOP 1981). Also by popular request, the animated graphics display you may have seen in Heathkit Electronic Centers is also included and has been modified for CP/M. Either version is \$29.95.

Using TOUCH TYPIST, you progress at your own pace given drill patterns to practice. There are three series of lessons. In the T series, you will learn to touch type on the standard keyboard, one letter at a time. In the N series, you will learn to touch type on the numeric keypad of the H19 or H89. In the S series, you will learn to increase your speed and accuracy on the standard keyboard.

In each lesson you will be given some practice drills to improve your accuracy. TOUCH TYPIST will display a line of text in the middle of the screen and instructions at the top of the screen. All you have to do is type the characters that you see in the middle of the screen. If you hit the correct key, nothing will happen. If you make a mistake, a large 'X' will be displayed under the incorrect letter.

Also available from NEWLINE SOFTWARE is the popular screen editor, VIDEO SCRIBE for \$39.95; and an H19 graphics editor, VIDEO ARTIST PLUS for \$39.95, both only for HDOS at this time.

GOODIES FROM FAR FIELD SOFTWARE

FAR FIELD SOFTWARE, 1031 Rue Verand, Slidell, LA 70458 has developed software for the HA-8-3 color graphics board and for various plotters and digitizers. "We at FAR FIELD have been using text processing software to develop program documentation, write letters etc. We don't know about the rest of you, but we find our technical staff to be outstanding computer engineers but lousy spellers(AMEN!--ed).

--Because of this, they have developed some word analysis programs, SPELL-GOOD and SUPERFOG. SPELL-GOOD reads a 3000 word dictionary, structures it, then reads the manuscript, looks for words not in the dictionary, lists in alphabetical order words not in the dictionary, and provides the option of adding the word not found to create a new dictionary for future use. SUPERFOG reads a manuscript and provides a detailed analysis on the number of words, the number of words per sentence, number of sentences, etc. But the most important calculation performed is a modified Gunning Fog Index (G) based on the average sentence length (S) and the percentage of words (W) with three or more syllables. It approximates the grade level of education needed to understand the manuscript. From there you can go back and simplify or modify to make your material more "readable".

SPELL-GOOD sells for \$39.95 available under HDOS or CP/M 2.2, and requires 48K. SUPERFOG sells for \$29.95 also available under HDOS or CP/M and requires 32K. Either is supplied on standard 5.25" or 8" discs. If you want to purchase both, they are available in a package for \$59.95. Add \$2 shipping for 5.25" discs, and \$4 for 8" discs.

RE-CONDITIONED KEYBOARD DIABLO PRINTERS

Phil Hoppa writes in to inform us that DTC (Date Terminals & Communications) 590 Division St, Campbell, CA 95008 (408) 378-1112 has available for \$1850 re-conditioned Diablo printers complete with keyboards. That's all I have on this, if you want to talk to Phil, write him at POB 50397, Palo Alto, CA 94303.

UPDATE ON PRODUCTS FROM M.I.-8

Ted Benglen of M.I.-8, 822 E. County Rd. 30, Ft. Collins, CO 80525 (303) 669-4116, writes to tell of the product line they

carry. His small business has become so successful thanks to the Heath public, that he has been able to quit his job and do "his thing" full time now. That means more support for you, and better service.

The hottest items going now are the new V-8 Model II speech synthesizer and the BSR Home Control Interface. For those who are relatively new subscribers, Ted was the pioneer in speech synthesis for the H8 computer. When I got my first evaluation unit quite some time back, I was very impressed and surprised. Now thanks to your support and trust, Ted has been able to devote full time in this field, to bring you even more and better products!

The BSR Interface offered by M.I.-8 is a modified version of the circuit originally sold by Micro-Mint for \$119. The M.I.-8 version is only 49.95 kit and \$79.95 assembled. The BSR Home Control Interface is an inexpensive way to try your hand at putting the computer to work controlling your environment. Because the BSR Interface uses parallel interface, the PC-12 (by Don Long--also a big item in past issues of H8SCOOP), clock parallel interface board is a natural. Controlling lights, appliances, and other household items in real time and temperature control was born with the mating of the BSR and PC-12. Now add V-8 Voice Synthesizer to announce time, temperature and current status of your home and you have moved right into the year 2001.

The PC-12 clock/parallel board for the H8 is a regular stocked item and can be purchased at a discount when purchased with any of M.I.-8's other products. And now, M.I.-8 accepts VISA and MC, along with COD's for phone orders.

Ted tells me because most of M.I.-8's products are designed for the H8 (the original idea behind H8SCOOP) he will make special offers to H8 owners and present the information exclusively in H8SCOOP! Another good reason to stay tuned to H8SCOOP--CONTACT WITH THE FUTURE!

The first offering will be the BSR Home Control Interface to readers of H8SCOOP (who must mention they saw it here when they place their orders to qualify for the discount--(woe be if any names appear on that list that are not subscribers!--ed)). The normal offer of \$79.95 for the assembled BSR interface will be offered for the kit price of \$49.95. That's right folks, assembled and tested board for the kit price--You saw it here first! Don't forget for all orders to include \$2.50 for shipping and insurance. This offer will extend for two months, through October only.

The BSR Interface board comes with owners manuals and sample running program. It can also be configured to plug directly into the H-8 buss for those not having a parallel port, although its best potential is realized when operated via parallel port. This board can also be connected to the H89 line via a standard parallel port.

The V8 Voice Synthesizer for \$349.95 assembled and tested, \$249.95 kit, comes with complete documentation and demo/diagnostic program on disc. There is more software coming soon for more applications of this device. This synthesizer has an unlimited vocabulary, and the best part is, it's easy to program. Several foreign languages are also presently being worked on for it. It comes with several demo programs, ranging from talking terminal to games and utilities. It has built in automatic inflection, for more natural sounds, which are clear, highly intelligible, and easy to understand. It comes with a parallel I/O and its own built in audio amplifier. Plug it into a parallel port and it's ready to go. Heath H8-2, or the PC-12 can be used with the H8, or connection to the H89's via user expandable I/O. This one's really good, write to them for info on it, and a demo tape. (Add \$10 for shipping and insurance for the V8.)

The PC-12 for the H8, to give you a battery backup on board programmable clock, PLUS 12 fully programmable parallel I/O ports is not \$695. No, not even \$595, but a measly \$264.95 (\$5 shipping and insurance) for the fully assembled and tested board with all parts, and clock set and support disc. The unit comes populated for 3 ports, add \$25 for each

additional 3. The clock? Reads seconds, minutes, hours, days, months, and years and supports interrupts.

MODEM KIT for \$74.95 (\$2.50 shipping and insurance) can be configured for answer or originate and be acoustic or direct coupled. Auto detect is available, and it can be wired for half or full duplex--all you need (beside software and money) to get on the HUG.BB. Easy to assemble and test.

And for the poor man who wants speech, for \$149.95 assembled and tested, or \$119.95 kit (\$2.50 shipping and insurance), you can have the H8/H89 DIGITALKER. It contains a built in vocabulary of 144 expressions. Many hundreds of words and phrases can be programmed, and thousands of new phrases can be added by changing ROM chips. You simply send an address to a port, and it speaks that contents (eg OUT 127,0 says "THIS IS DIGITALKER"). It connects to the H8 buss, H8-2 parallel card, PC-12 parallel card, or H89 via I/O expansion board.

MORE FROM W.I.S.E.

W.I.S.E./422 Third St/Baraboo, WI 53913/(608) 356-9432 adds a serial interface board for the EPSON printer. Refer to AUGUST 1891 H8SCOOP, Page 6 for information on their serial/parallel/clock board for the H89. This serial interface board is a nonbuffered board that allows the EPSON printers to accept serial RE-232 or TTL data. There is no current loop or buffers on the board. Baud rate, stop bits, and parity is programmable with jumpers. With handshaking, a baud rate of 9600 can be attained. The serial interface board will sell for \$37.50 and should be available in September. This board fits inside the EPSON, and if you buy it from EPSON, costs about double this price.

NEW SPOOLER FOR HDOS 2.0

The EASYSOFT DESPOOLER for HDOS 2.0 for \$39.95 is the best Spooler I have seen around as of this point. It is available only from Tom Jorgenson, 122 Yankee Drive, St. Charles, MO 63301 (314) 946-1968 after 4:30PM CDT.

So what's a spooler? A spooler allows you to be printing a file while your computer is doing something else, a sort of time share arrangement. The EasySoft Despooler is a package of device drivers which when called via a PIP copy command or from another language will perform as a background file printer.

This means the file will continue being printed even though the user is still operating his system normally performing other commands.

Five device drivers are supplied, four drivers replace the standard HDOS distribution drivers (LPH14, LPH24, LPH44, and LPH84) completely, the fifth driver will call and use a user-supplied LP: driver to perform printing and formatting of the despoiled text--a very versatile spooler indeed! These drivers are very fast, operating an H14 at 4800 baud and an H19 at 9600 baud yields no apparent delay except during the short read/write time when sectors are being moved between the disk and computer memory.

The driver SET options are all identical to standard Heath drivers with the one exception that the SPH84 driver now supports pagination which may be disabled. The Diablo, SPH44 driver prints BI-DIRECTIONALLY as well!

No strange installation procedures are involved, the setup is the same as with any other device driver. No disk areas are used, no disk writes are performed by the driver, no HDOS mods or "tricky" programming. All the internal code is performed through normal HDOS procedures. No channels are tied up by the driver either.

The drivers maintain a queue stack of 6 file names, when full the stack will clear out the names as the files are printed. These drivers are callable from any language capable of reading via channel 3, and a Microsoft Basic demonstration file is supplied as an example. The package also contains an

ABORT utility to immediately halt operation of the despooler when necessary.

I tried the despooler with the new SY.DVD device driver and the 400K Tandon drives and it worked with the new driver, but only on the old 100K drives. You cannot copy a file from the 400K to be spooled out, unless you go to the 100K drive first. I highly recommend the spooler to anyone wanting to make the most of their system, i.e., print files and not tie up the computer doing it!

Tom Jorgenson, is the author of THE PARALLELER, distributed only by Keyboard Studio, as mentioned on page 5 of AUGUST H8SCOOP. Tom is also working on a special device driver for the TYPE 'N' TALK sold through the Keyboard Studio. He is also currently working on an HDOS bulletin board package using THE PARALLELER and an auto answer modem. The telephone number of the St. Louis BB using this package as reported in last month's H8SCOOP (page 5) has been changed to (314) 291-1854. If you want to get a demonstration of this package, call the number and experiment with the BB.

UPDATE FROM KEYBOARD STUDIO

From The Keyboard Studio, 125 Aspen, Birmingham, MI 48009 (313) 645-5365 comes some software and hardware items of interest I did not have a chance to print up properly last month. A software goodie a lot of you have been asking for--a program to help keep track of your magazine articles, etc. I'll let Ray tell it. "First, a software product called KEY-IT! This is a new Keyword in Context (KWIC) program for the H89 or H8/H19. How many times have you thumbed through old magazines to look for an article or articles on a specific topic? Probably quite a few, but with KEY-IT! you won't have to again. KEY-IT! will generate a cross-reference of magazine articles, books, periodicals, files or just about anything you like. You can select hardcopy, video or disk file output on related Titles, Author's names, Subject matter, or any other related keyword that you desire! This could save you literally hours of searching. KEY-IT! uses a windowing effect when the output is to the video console. It is available for \$24.95 (HDOS) or \$26.95 (CP/M) and requires at least 48K of memory. It can be purchased direct from us or at most of the Heathkit stores around the country.

Another product to make your computer talk is the "TYPE-N-TALK" by Vortax. This unit is compact, and has many special programming modes, selectable baud rate and GREAT sounding speech with automatic inflection. Its vocabulary is unlimited. To make this unit say a word or sentences, you simply output the text to a device driver just as you would with a printer. For example, PRINT #1, "HELLO MY NAME IS VORTAX" will speak just those words automatically. It is supplied with device driver software patch, and a cable connection diagram needed to plug it in to any RS-232 serial port. The Keyboard Studio discount price to you is \$339, PPD (lists for \$375).

The TYPE-N-TALK comes complete with built in audio amplifier, just add a speaker and away you go. It has frequency control (for controlling speed of talking (and actually pitch) and a volume control. If you want a demo cassette tape of the sounds, send \$3.00.

The Keyboard Studio is also a distributor for The Information Center SOURCE BOOK, a compilation of many sources for software, hardware, printed matter, services and dealers all devoted to Heath/Zenith computers. Details are given for many products that are currently available both hardware and software. It contains over 150 pages in a loose leaf binder with plastic tab divisions. A complete source of who's got what--an excellent referent to have around.

DEVICE DRIVERS FROM GRANT GUSTAFSON

Various device drivers are available from Grant Gustafson/2243 S. Belaire Dr/SLC, UT 84109 (801) 581-6879. These newly assembled drivers utilize all features that the manufacturers advertise. The disk for each includes DOC,

LP.DVD, LP.ASM, cable information, interface information for HDOS and CP/M. These drivers run under HDOS with NO hardware modifications. Interfacing amounts to a cable (and an off-the-shelf interface board for the MX-80). The pricing on these will be about \$19.95 each, contact Grant for more info on this. Presently drivers are available for the Anadex, Epson, and Comet printers.

The ANADEX drivers for the Models 9500 and 9501 has SET options for Baud, Columns, Length, LPI, Page, Port, Skip, Width, Auto Line Feed, Form Feed, Graphics mode, and Wide print mode. It is a very versatile device driver written specifically for this printer to take full advantage of its capabilities.

The COMET drivers allow running under HDOS with bi-directional print at 9600 baud. SET options include Baud, Margin, Page, Port, Width, Condensed column, Form Feed, VTAB, and WIDE. The Normal print of 80 columns prints at 125 lines per minute! Condensed print of 132 columns is available which prints slower, Medium Expanded print of 66 columns, and Large Expanded Print of 40 columns. Left and Right margins can be set for any values.

The most popular printer now is the EPSON MX-80 because of its low cost and versatility. The EPSON device driver includes automatic configuration of the printer for various print sizes and fonts of the EPSON. The driver takes care of printer initialization, line and page counts, tabs, and font changes at the front of a line. The SET options include BUAD, Length, LPI, Page, Port, Width, **BOLD OVERSTRIKE**, condensed 132 column print, wide letters, Letter quality print, and Form Feed. Ample documentation is supplied to allow the user to be able to set these features in line, from a BASIC program for instance, by sending CHR\$() characters. The driver is written to make maximum use of the features of the ever popular Epson, and yet for a reasonable price.

For more information on these, and exact pricing, contact Grant.

Grant tells me he will be coming out with more software soon. Included in these to be DIAL.ABS--15 sector modem program for the H8/H8-4 or H89. It will support standard and auto-answer modems. CRASH.ABS is a dead disc save utility. SND.ABS and GET.ABS is a high speed H89-H89 (Or H8/H8-4) modem file transfer. DSKBATCH, TAPBATCHM, DISKTAPE is an H89 tape archival utilities for standard cassette backing disc (also for H8). PIRATE.ABS, COMP.ABS, and SNOOP.ABS are 5" disc and file compare programs. They are useful to document patches in DUMP.ABS formats, or for finding any differences in a disc from another. I'll keep you in touch as I hear more on these.

FBE CENTRONICS 737 MOD

Daniel Morin, 124 Plymouth St, Manchester, NH 03102 has completed a software modification to FBE Researches' Centronics 737.DVD. The mod allows tabs to be processed in the proportional space mode of operation. Now in all three character fonts the DVD aligns the tabs so they are in the same format as the H19-H8/H89 tabs are. This is available from Daniel for \$3 and the original FBE.DVD disk which will be returned with his CT.DVD and MCT.ASM files added to it. This cost covers shipping and handling.

UPDATE FROM TRIONYX

The new Z80 CPU for the H8 is on schedule for late September release, available as bare board for \$75, kit form for \$200, and fully assembled for \$250. Advanced orders will be accepted.

The new board will be very easy to install. Only a simple mod to the system is required. The CPU clock rate may be changed at any time using either system or program software. It can also be changed during program execution using PEEK and POKE routines with a processor status port (PSP). The Z80 CPU board has heavy duty bus drivers and a front end bus termination network for reliable operation. The Heath 0-org circuits are incorporated on the board, eliminating the need

for the separate Heath Org Zero card.

Single step instructions may be executed from the front panel. Single or multiple wait states (or none) may be inserted at 4 Mhz. No wait states are used at 2 Mhz. An extra socket for an additional ROM or RAM chip has also been included.

Because there wasn't room on the board to include a clock circuit or arithmetic chip circuit, they will be packaged individually on "half boards", similar in concept to the Heath 0-org board. They are designed to plug into the T-H90 motherboard in to auxiliary positions, or into the standard Heath motherboard slots.

A large number of additional half boards are planned, including memory bank select port, EPROM card, A/D card, D/A card, Voice Synthesis card, Appliance Control card, Battery Backup card, and the Mass Storage Peripheral controller.

They also tell me the new T-H90 motherboard has been very well received. Their tests show that the center ground layer by itself provides significant improvement in the signal transmission characteristics of the bus. In most cases the extra ground connections provided on the B connectors will not be needed.

QUIKDATA TO DISTRIBUTE SPELLBINDER

I have been so impressed with SPELLBINDER I have decided to sell it for the H8-H19/H89 systems. Hopefully as of next H8SCOOP I will be all set. It is not sold as a word processor system but as an Office Management system, because of its power and features such as sort by fields, merge, word processing, etc. The list price is \$495 but I hope to be offering it for about \$295 (tentatively). I'll have the rest of the details next month. Anyone wishing some info on it, contact me.

* PROGRAMMING GOODIES *

CP/M & HDOS PROGRAMS ON SAME DISC

The October issue of REMark will be carrying an article on how to have both CP/M and HDOS programs on the same disc. I will not go into detail here, but list the program to do it for those of you that want to get a head start. This was given to me and written by Bob Ellerton of HUG. There are lot of uses for this, the best is for those who distribute the same programs and utilities in both HDOS and CP/M format. Now this could be done on one disc instead having to send out two.

The concept behind this is simple, it's a wonder why it wasn't thought of before. Write E5's in the first 114 sectors of a disc for CP/M, and use the rest for HDOS. E5 is CPM format. Since the HDOS directory starts at sector 120, we're safe. Track 3 is the CP/M directory, and track 13 is HDOS, so there're no conflicts. If you were to go all the way with writing the E5's instead of just 114 sectors, the entire disc would then be formatted for CP/M.

```
10 CLEAR 300
20 OPEN "O",1,"SY1:CPM.SPC"
30 FOR I=1 TO 114
40 PRINT #1, STRING$(255,&HE5)
50 NEXT I
60 CLOSE
```

First INIT a fresh disk under HDOS, and then run the format program, this one listed under MBASIC. The 114 sectors will hold about 20K worth of program so be sure not to exceed that. The CP/M programs can be PIPed onto the disc, and HDOS programs can be PIPed as well. Be certain when you're done to set SLW Flags on the CPM.SPC file so HDOS doesn't try to play with it, since the file name will reside in HDOS directory, but will never be used.

For a full discussion on this, read the OCTOBER REMark. 6

Many thanks Bob.

DIABLO FORMAT QUICKIE

From Craig Todtschinder of the National Software Register comes this quicky for setting lines per page. First you must set #lines per inch then lines per page in the driver, then SET PAGE LENGTH CHR\$(27);CHR\$(12);CHR\$(#); where # is the # lines per page you want.

REQUESTS

James D. Smith (JD), 338 West Mill St, Bonduel, WI 54107 (715) 758-2188 wants this thrown out to H8SCOOPers. Anyone with an idea, contact James. "A couple of wierd things have happened to me while using HDOS 2.0. Once, after typing CAT SY0:/B (cr), HDOS crashed, the screen cleared and the display for the rotational speed check came on, but without running the text itself. And of the two DUP programs that I have, one doesn't work at all the the other works when it wants to. That applies as well as when I've booted up with version 1.6 and run them, as when I've booted up with version 2.0. I can understand a program written to run under 1.6 not working with 2.0, but why won't it copy a version 2.0 initialized disk when version 1.6 is allegedly in control? Are some things left in memory from a previous boot screwing up the works somehow?"

From Tom Ehlert, 8118 Jackson Pk. Blvd, Wauwatosa, WI 53213 (414) 771-5346 "Is there a smart person out there who knows how to trick a Heath H14 line printer into accepting software commands to cause the 0.020 inch paper feeds forward and reverse which are normally obtained with the FEED FWD and FEED REV front panel switches?" Any got the SCOOP on that, Tom would appreciate it.

More on printers, this time the DIABLO by Lloyd Johnson, President Leejon Engineering Co, 700 Highview Rd, East Peoria IL 61611 (309) 699-1476. "You may have seen an article I wrote in REMark, issue 13, on DIABLO REMOTE CONTROL. I have found it more convenient to embed control codes in the text of letters, such as this, for direct control of my printer than to use a text format program. By doing so I no longer need to load a program to print copies. The only shortcoming I've encountered is inability, so far, to enter a code byte which will stop the printer at the end of a page to allow me to change paper. To write letters or articles of more than one page requires opening of separate files for each. Incidentally I use PIE v. 1.5 to create text files as with it CTRL-n codes can be inserted into the file. Unfortunately CTRL-S doesn't stop the printer".

If anyone has worked out a patch to LP.DVD or some other scheme which would provide the stop printer capability, please contact Lloyd.

George Duke, 8350 SW 184 TER, Miami, FL 33157 sent in this open letter to be printed in H8SCOOP. Anyone having any leads on the mysterious disappearance, please contact George.

Open letter to "Metor-Detroit area Heath Computer Users Group"
Dear Chuck Dattolo, Ray Klatt, Clyde Hipwell, Neil Coffin, Et Al:

In January of 1980 I took advantage of your kind offer to let nonmember fellow Heath Users subscribe to your club newsletter "THE DUMP". I found it a very enjoyable publication and looked forward to each issue. In your October 1980 issue you announced some interesting articles to come and also reminded us that it was time to think about renewing subscriptions. You revealed that the \$3.00 fee to nonmembers did not cover the cost of mailing but you were glad to do it and share your fine publication with others.

I promptly sent you a check (for more than the requested amount to help you in the postage dept.) to renew my subscription. That was the last I heard from you!!! My subscription stopped abruptly with the October 1980 issue

without even finishing up my first year subscription!!! You cashed my check so I know you recieved my letter. The money is not important but if you continued printing "THE DUMP" I am dying to find out what interesting things appear therein!!

Since I have written several followup letters with no response, I feel sure that you were all beamed off the face of the earth by some alien power during a secret club meeting. If anyone out there witnessed their disappearance would you please let me know as they (The Dump) are sorely missed."

No, they were not ALL beamed off. OK Clyde, I checked my subscriber list and you are within!! Get in touch with George, and give him the SCOOP! What indeed did happen?

From Gray Hoffman, 1502 Spillers Ln, Houston, TX 77043-- "Does anyone have information on a circuit for the H89 that would output composite video and thus allow a remote monitor to echo the computer display?" If anyone has this, please direct it to H8SCOOP also, as I would like to print it for the benefit of all.

From WILLIAM GOODWIN, 312 Lamson, Grand Ledge, MI 48837. "I am interested in CP/M. I have obtained a copy of the CP/M Users Group Catalog and it is an amazing list of software! Unfortunately, the only 5" disks they handle are North Star, Double Density. If a double density controller becomes available for the H8 and your drives, (the Tandon--ed) will the resulting system be able to read North Star CP/M discs? If it won't work this way, perhaps you can suggest that someone with 8" drives might be able to pick up some extra money transferring material from the CP/MUG 8-inch disks to Heath 5" disks for sale.

"Another way to obtain CP/MUG material would be a controller for the H-8 that could handle a single one-sided single density CP/M compatible drive. Such a controller should be rather inexpensive. This would permit an H8 to read CP/MUG 8" disks."

Let me say, don't over look the possibility of transferring via MODEM, even hard wire between a machine with North Star compatible or 8" with H8. If you have access to such a setup, it would be the easiest way.

I transfered this message from the HUG.BB, from Tom Teimpidis on 13-Aug-81. It concerns 8" for H8, so read on.

"Ray Livingston, 2386-323 E. Del Mar Blvd, Pasadena, CA 91107, 213-792-4763 after 5PM Pacific Time has in the works a very low cost controller for the H8 that will support 8" drives, either single or double sided but not double density. You must be using a Z80 CPU however as it uses the non-maskable interrupt capability. He is planning to offer instructions, schematics, software, a new PROM, etc. It is easy to wire wrap, using only about 14 IC's. If there is enough interest he will offer a P.C board and perhaps finished boards, etc.

"This is a very low cost way to have 8" capacity and IBM 3740 CP/M compatability. Anyone interested, contact Ray." The next message indicated the controller "uses standard "DUMB" Shugart compatible drives and is complete CP/M compabable." Some of you may want to investigate this route, as it seems to be something for the H8 right around the corner.

From Georges Troupin, 22 Rue De Grady, 4920 - EMBOURG, BELGIUM. "I got the Extended Configuration Option for the H8 (To run CP/M org 0), really no problem at all, it worked as soon as it was installed in any position of the H8 bus. But my former 56K system lost 8K when running HDOS. I tried the HDOS 1.6 Patches for 64K RAM, published in BUSS #32 but without any success for both of them. If anyone has a good patch for HDOS 1.6 and HDOS 2.0 to recover the 8K it would be very useful for a lot of us.

Van Christopher, 9602 Pebble Beach Dr, Santee, CA 92071 is interested in converting Radio Shack Basic to BH Extended

BASIC for converting programs found in magazines etc, to run on our systems. He is especially curious about the Radio Shack SET command--to convert it to BH Extended BASIC. He runs mostly the cassette software. Van, see July issue of Microcomputing, Page 92 on H19 graphics from TRS-80 by D.C. Shoemaker (this pointed out to my be Brother Monroe) Any info on that one, contact Van.

W H O ' S W H O

MICROS FOR USE WITH HANDICAPPED

John Bennen, 620 Hill Street, West Baraboo, WI 53913 is interested in corresponding with anyone developing uses of micros with the handicapped. He has been working with this idea to help the handicapped. The Headlines from February 23, 1981 WISCONSIN STATE JOURNAL reads "Cerebral palsy victim's computer helps bridge communications gap". It's about a 17 year old stricken with cerebral palsy, confined to a wheelchair, and can only control one arm and one finger.

John Bennin, School psychologist developed some concepts on his H8 and converted them to the school APPLE to help Joe. For the first time in his life, Joe can communicate quickly and accurately. Special programs were developed to allow Joe to press several keys to call us a list of 80 sentences he regularly needs. He can also choose from tables of words arranged alphabetically to compose a message. "What is now time consuming, was before an impossibility" comments Bennin. John Bennin's next project will be adding Voice to the computer, so when Joe types a message, it will say it and people won't have to come to the screen to read what Joe has typed.

If anyone has been working in this area, or is interested in this work, get in touch with John. He would also like GOOD CALL stuff which is very hard to come by and would be interested in corresponding with readers having general interests in this field as well.

I feel this is one of the better, unselfish uses of a computer. If you can help out John, get in touch with him.

GOOD GUYS and BAD GUYS

SUPER BAD GUY--SIRIUS SYSTEMS. Don Texeira wrote in some interesting information about Sirius Systems that cost him over \$250. I seems most of the major microcomputing magazines have been advertizing this place for such items as disc drives. Well Don sent them \$266 for an MPI 51 drive and never received it although they cashed his check. Writing and phone calls never got an answer or response. Well after much investigation by the Postal Service for mail fraud, magazines, and many government agencies, it turns out they went Bankrupt. This means Don, and many like him will never see their money again, nor receive the product. So stay far away from this place and others like it. I try to devote a good part of my H8SCOOP time to the Good and Bad guys column in hopes things like this will never happen. Still, the column is too often neglected much to the sorrow of my subscribers. Read and follow the column. If you see them in GOOD GUYS, be assured they are doing a good job. If you find them in BAD GUYS, stay away from them. By all means, if you have a good guy or bad guy, write in and let me know, so we can prevent this from happening to anyone else, and so you can get the best service, price and deals around.

BAD GUYS--CPU SHOP in Charleston, MA. See Issue #'s 17- page 8, 13-page 11. I've had more bad reports from them, I cannot recommend them even though I received a letter commending them (out of a good dozen saying stay away). Brian Hansen sends me this update from last month.

"I received my printer 8-3-81, on 6-24-81 they had promised two weeks to ship. On 7-23-81 I again complained and received a prompt response I was put in a priority file and the unit shipped 7-31-81. I had also sent a letter directly to

EPSON about the CPU shop which might have had pushed things faster since they sent a copy of their response to the CPU shop. I realize that the back ordered unit wasn't the CPU shop's fault. I don't feel they kept me properly informed and I still can't recommend them."

CLASSIFIEDS

FOR SALE--H8-5 Serial/Cassette board with cables and documentation, \$70. ECP3801A recorder, \$30, or both for \$95. Ted Guest, 4366 Carlo Dr, Dayton, OH 45429. Phone (513) 298-6084.

FOR SALE--H8 computer with 2-8K memory boards, 1 H8-5 serial board with cables and cassette player, H9 terminal, and H17 single drive disk system with controller board. All this is virtually new. Will consider selling separately, but would prefer to sell the entire package for the best offered price. Harry Albaun, 108 Seaman Av, Rockville Ctr, NY 11570. Phone (516) 704-4690.

WILL TRADE--H8-2 parallel board for H8-4 four port serial board. Will sell H9 terminal, has lower to upper case mod, excellent condition, \$225. Jim Bates, 10610 Sageburrow, Houston, TX 77089. Phone (713) 944-4687.

FOR SALE--Siemens FDD 100-5B 5.25" disk drive. Brand new, for Heath equipment, \$245. Jerry Phelps, 6013 Innes Trace Rd, Louisville, KY 40222. Phone (502) 425-4765.

FOR SALE--Applied Digital Data Systems (ADDS) Regent 100 CRT with options. Like new condition. \$700 or willing to trade for H19. Tom Roche, 7 Winkel Way, Ballston Lake, NY 12019. Phone (518) 877-5189 evenings.

FOR SALE--Mullen extender board for H8, \$25; A/D converter board from CCM Inc, never used, original \$125 asking \$60 or offer; Mullen Computer products breadboard for H8 never used \$25. Chuck Ballinger, RFD 1 Box 676, Spokane, WA 99204. Phone (509) 448-9727 or Micronet 70235,313.

FOR SALE--32K DG Electronics Dynamic RAM board fully populated, manual included. \$250 includes shipping. Also Heath H-14 line printer fully assembled. \$400 includes all manuals and shipping. Tom Crittenden, 4428 Scarborough Dr, Baton Rouge, LA 70814. Phone (504) 272-0674.

FOR SALE--Complete Heathkit H8 based 32K system. Includes H9 terminal, H17 dual disk drives (204.8K), H8-5 Serial I/O card, H8-4 parallel I/O card(??-ed), H8-7 interface card, and complete system software including BASIC, ASM, EDIT, DEBUG. Full documentation provided. \$1500. Dr. J. Trivisonno, John Carrol University, Cleveland, OH 44118. Phone (216) 491-4301.

WANTED--H9 with upper and lower case mods and 24 line mod installed. Also wanted is an H17 single drive disk system with controller board. Will buy or trade for any of M.I.-8's products (see NEW STUFF). Contact Ted Benglen of M.I.-8, 822 E. County Rd. 30, Ft. Collins, CO 80525. Phone (303) 669-4116.

WANTED--Z89 computers at a reasonable cost. Contact Allen Bernson, POB 68, Everett, MA 02167.

PEEKING & POKING

THE BIG THREE

The more I read other newsletters and talk to their editors, the more I am realizing we all have to work together, and yes, we all need each other. I'm mainly speaking of the big 3, in order of circulation REMark, BUSS and H8SCOOP. When I started H8SCOOP I did not start it with the intention of running REMark or BUSS out of business, but to give you something different and extra, to compliment the others. I figured the wise and serious computer individual will already be subscribing to REMark and BUSS.

Although some similar articles pop up in sometimes two or three of these newsletters, I try to discourage it knowing that most of you do subscribe to more than one. If you have been following H8SCOOP you know what my policy has been on that. The fact is, that sometimes it does happen and it is not avoidable since we (the editors) have no idea what has been submitted to the others and what they are going to print!

The format of H8SCOOP is totally different than that of BUSS, who is also totally different than that of REMark. They all have their places. While BUSS and H8SCOOP are more current than REMark, REMark has more long articles and general interest articles. This can be understood when you look at the average length of REMark being 32 pages and the others 8 pages. Because REMark has the glossier pages and slick format it takes longer to print. Does that mean nix on REMark? No way. It has its place and has been steadily improving. In fact they all have been steadily improving. That's a good sign because it means they are all getting your support, and they are all providing support to make the Heathkit line of computers more useful and popular.

I feel it would be a major loss to lose any of THE BIG THREE at this point and I advocate your support of them by subscribing and sending in articles, hints and information. I would not give up my subscription to REMark or BUSS for anything--they're too valuable to me. If you like them, then support them. Its always amusing that the ones that give least or no support are the ones that complain the most. This point was explained in a past H8SCOOP about my "friend" who points out and criticizes my spelling--He's not even a subscriber!

The more we stick together and help each other, the better we'll all be.

Henry Fale

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Subscription Rates--\$15/Year (\$21 overseas AIRMAIL)--current issues
\$12/Year (\$18 overseas AIRMAIL)--first year back issues.

Back issues #'s 1, 2, 7, 8, 9, 10 no longer available. Remainder through March 1981 still available for \$1 each while supply lasts.

PHONE HOURS: Almost anytime you can catch me except sunset Friday to sunset Saturday. NO COLLECT CALLS